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Two birds with one stone: Query-dependent moment retrieval in muted video or audio via inter-token interactions

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ABSTRACT

The surge of social media has led to massive growth in audio and video content, increasing the demand for tools that retrieve user-specific highlights, which are short segments aligned with sentence queries. In real-world scenarios, users often access only one modality, such as audio companions or muted video mode, and may switch between them depending on context. However, current methods treat muted video and audio retrieval as separate tasks, lacking a unified structure to support flexible modality adaptation. We propose Inter-token Aware Retrieval (IAR), a lightweight and unified framework for query-dependent moment retrieval under unimodal constraints. IAR models audio/muted video and text as token sequences, and captures both intra-modal and cross-modal token interactions. It consists of three modules: a Boundary Enhancement module that strengthens token-level contrast between event and background regions, a Multi-Modal Alignment module that enhances token-to-token relevance between the query and a single modality (video/audio), and a Context Convolution module that aggregates local token relationships while preserving temporal continuity. A two-stage 2D map is used to score and select the best-matching moment. IAR achieves strong performance on Charades-STA, ActivityNet-Captions, and AudioGrounding-AMR, and also demonstrates effective generalization across muted video and audio tasks with high computational efficiency.

1. Introduction

In recent years, online media platforms have seen an explosion, with at least 500 hours of video being uploaded to YouTube every minute[2] and 74,000 audiobooks being released annually[3]. Thanks to the low production cost and high sharing efficiency, they attract vast audiences. However, this unprecedented expansion necessitates accelerated user-preferred localization given attentional limits [1–3]. Therefore, it is practical and beneficial to supply a unique moment according to user-specific preferences [1,4]. For example, UMT [4] can handle different combinations of input modalities based on user queries and output key moments. This underscores the centrality of query-dependent moment retrieval, where the system identifies highlights relative to a specific query rather than general content.

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[2] https://www.statista.com/statistics/259477/hours-of-video-uploaded-to-youtube-every-minute/

[3] https://www.publishersweekly.com/pw/by-topic/industry-news/audio-books/article/89547-audiobook-growth-continues.html

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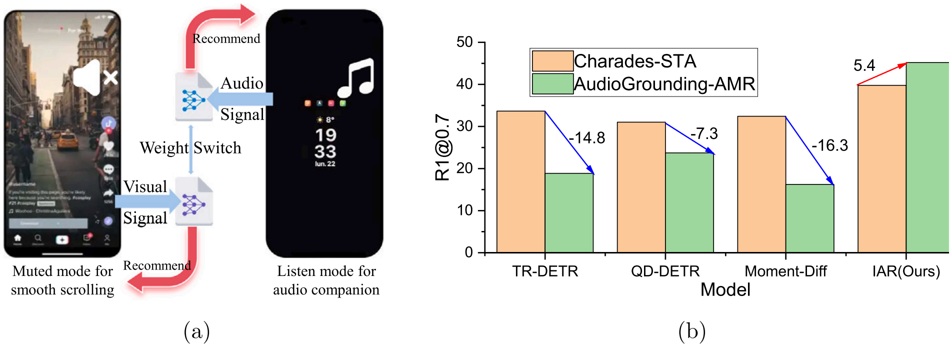


Fig. 1. (a) An example application of task switching between muted video and audio. (b) Performance comparison on Charades-STA (VMR) and AudioGrounding-AMR (AMR). Existing models show performance drops on AMR, while our IAR model improves by 5.4, highlighting better cross-task generalization.

While most existing research focuses on Muted Video Moment Retrieval (MVMR), where only visual features are involved [5], Audio Moment Retrieval (AMR) remains relatively underexplored. In many practical scenarios, users, particularly those with visual or auditory impairments, often have access to only a single modality [6]. Moreover, user-preferred content localization usually requires switching between modalities. For instance, in social media feeds (e.g., TikTok, Instagram Reels), muted autoplay with visual-only presentation is standard to capture attention without impeding scrolling velocity. Conversely, during audio companion mode or under low-bandwidth conditions, users predominantly activate background audio mode, relying exclusively on auditory information (Fig. 1a). The dominant research paradigm treats MVMR and AMR as separate tasks, with isolated model designs and training procedures. *This task-level fragmentation leads to inefficiencies in deployment and limits the system's ability to switch between modalities flexibly*.

MVMR and AMR share significant similarities in logic and structure. MVMR localizes events by sentence queries in videos [7,8], and AMR analogously for audio [9]. Although existing models demonstrate prominent performance [8,9], feature heterogeneity between video and audio modalities impedes single-model adaptation across dual tasks. Despite well-established MVMR research, directly adapting MVMR models to AMR datasets exhibits sharp performance degradation (Fig. 1b). A plausible explanation is that excessive focus on contextual information during video/audio-query matching diminishes temporal feature prioritization, blurs boundary detection, and hinders retrieval performance [10].

To meet the growing demand for flexible moment retrieval under uni-modal constraints, where only one modality is available at inference time, we aim to design algorithms that can handle either muted video or audio inputs—not simultaneously, but within a unified model structure. This motivates the development of a modality-adaptive architecture capable of task-level switching between MVMR and AMR. Prior approaches often rely heavily on contextual semantics for matching video/audio with queries, which can blur temporal boundaries and dilute the distinctiveness of salient segments. To address this, we construct our retrieval framework from a token-level perspective, explicitly modeling inter-token interactions to preserve temporal independence and enhance boundary sensitivity.

To this end, we propose **I**nter-token **A**ware **R**etrieval (**IAR**), a unified framework that models intra- and cross-modal token interactions. Specifically, muted video/audio and text are divided into tokens, each treated as a node. We identify two types of interactions between node pairs: uni-modal (within video or audio) and multi-modal (between video/audio and text). To leverage these, IAR incorporates three core modules: (1) Boundary Enhancement (BE) applies average feature injection to magnify contrast between event and background segments, thereby sharpening temporal boundary representations in uni-modal streams. (2) Multi-Modal Alignment (MMA) employs cross-modal attention to strengthen alignment between query tokens and salient visual or acoustic cues. (3) Context Convolution (CC) aggregates local token relationships using aligned features, while preserving the temporal continuity of the original uni-modal stream. By integrating these components, IAR implements a two-stage retrieval pipeline using a 2D temporal map for moment scoring.

Our IAR outperforms existing methods on both MVMR and AMR tasks. More importantly, unlike prior models that suffer significant performance degradation when transferred across modalities, IAR demonstrates strong generalization between muted video and audio retrieval—achieving a positive transfer of 5.4 R1@0.7 from Charades-STA [7] to self-constructed AudioGrounding-AMR, as illustrated in Fig. 1b. This highlights the effectiveness of our unified architecture in supporting modality-level task switching.

To sum up, our main contributions are as follows:

- We unify the experimental settings of MVMR and AMR by reconstructing the AMR dataset, enabling direct comparison under consistent protocols. This setting supports fair benchmarking and validates the cross-task generalization ability of our IAR framework.
- We propose a lightweight unified inter-token aware retrieval structure for MVMR and AMR. It jointly models uni-modal and cross-modal token interactions while preserving temporal boundary cues critical for moment localization.

- Extensive experiments on standard benchmarks demonstrate that our model not only achieves strong performance in both MVMR and AMR tasks, but also exhibits superior cross-task transferability, with improved retrieval accuracy even when switching modalities.

2. Related work

Muted Video Moment Retrieval. MVMR aims at locating a temporal moment according to a query sentence using visual features only [11,12]. It is a variant of Video Moment Retrieval (VMR), also known as Temporal Sentence Grounding in Video, where audio is optionally included as an auxiliary modality to enhance retrieval performance [5]. However, most existing VMR methods focus on vision-language feature interaction while exclusively utilizing visual features. Directly comparing audio-incorporated approaches with audio-excluded ones would be methodologically unsound. Given that our research scope is strictly confined to uni-modal constraints, *we designate methods employing solely visual features as MVMR for dedicated comparative analysis*. Formally, each training sample is a triplet: a query sentence, a muted video, and a time interval of the moment that matches the query sentence. Existing methods can be divided into one-stage and two-stage approaches, depending on whether a stage involves pre-generating candidate moments.

The two-stage methods first generate numerous candidate moments from a video and then select the best one from all candidate moments according to the query sentences [7]. It has two paradigms: anchor-based and 2D-based methods. Anchor-based methods pre-define some temporal intervals with different proportions for multi-modal candidates, such as TGN [13] and SCDM [14]. 2D-based models build a 2D map to obtain fine-grained multi-modal features explicitly modeling temporal relationships, such as 2D-TAN [15] and MS-2D-TAN [8]. The following methods combine the 2D map with other techniques, such as graph neural networks [16,17], capsule networks [18], pyramid structures [19], and contrastive learning [20].

The one-stage methods directly estimate the start and end positions corresponding to a sentence query without generating candidate moments [21]. It also has two paradigms: regression-based and span-based methods. Regression-based methods directly estimate start and end timestamps [21,22]. Span-based methods do not directly regress timestamps but instead predict the probability that each video snippet may be the start or end position of the target moment [23,24]. Recently, some methods have adopted large-scale pre-trained models as the feature encoder to improve performance [25–27]. In addition to supervised approaches, there has been growing research interest in weakly supervised [28] and unsupervised methods [29,30], driven by their potential to alleviate the burden of temporal and semantic annotation, which remains a critical bottleneck in large-scale video understanding tasks.

Although existing methods have achieved high performance, some critical research gaps remain understudied, such as inadequate cross-modal interaction [11,13,27], fine-grained semantic misalignment [12,14,22], incomplete temporal contextual modeling [15,17], and suboptimal computational efficiency [18,25]. A fundamental cause lies in the failure to preserve the intrinsic characteristics of individual modalities during fine-grained multi-modal interaction, which inevitably leads to semantic distortion and feature degradation.

Audio moment retrieval. Unlike MVMR, AMR aims to localize query-relevant temporal moments from audio-only streams. Since AMR is a relatively new task, there is no unified paradigm in place. Early efforts explored various directions: [31] investigated unsupervised learning; based on the weakly supervised learning, [32] introduced contrastive learning under weak supervision; and [33] incorporated both sentence-level and phrase-level features. Gradually, two paradigms have emerged. The first regards this task as highlight detection, but it also fails to meet individual preferences. The second views this task as query-based moment retrieval, aligned with MVMR in formulation. In this work, we follow the second paradigm and propose a unified framework, IAR, that unifies the experimental setups of AMR and MVMR, enabling effective task-level adaptation within a single model.

Graph neural network in moment retrieval. Graphs are structures explicitly modeling complex relations in many fields. Recent studies model moment-wise temporal relations with graphs, and the formulations have progressed to jointly represent both the content and the query [34]. Furthermore, [16] proposed a multi-modal graph matching framework. In addition to modeling multi-modal relationships, some methods also use graphs to identify candidate moments while introducing context [8]. However, in multi-modal fusion settings, over-integration of features can obscure modality-specific temporal cues and degrade boundary precision. In contrast, our IAR framework extends graph-based reasoning by explicitly preserving uni-modal temporal structure within the audio or muted video stream. This allows us to model cross-modal relationships while maintaining task-specific temporal boundaries, which are essential for accurate moment retrieval.

Multi-modal interaction in moment retrieval Multi-modal interaction is a critical component in moment retrieval, enabling models to align content and query semantics across modalities. Existing approaches adopt various mechanisms including conventional convolutional neural networks [15,35], graph neural networks [16,17], DETR-based networks [22,36], etc. In this work, we construct a multi-modal interaction graph using graph convolutional networks, where one stream represents the uni-modal content and the other the textual query. Notably, we introduce boundary-aware enhancement and token-wise alignment strategies that preserve uni-modal temporal properties during cross-modal interaction, supporting accurate retrieval even in modality-specific scenarios.

3. Proposed method

3.1. Overview

Given an untrimmed muted video V or audio A and a query sentence Q, an MVMR/AMR model f would localize a moment (τ_s, τ_e) starting from τ_s and ending at τ_e in V/A for Q:

$$(\tau_s, \tau_e) = f(M, Q), \tag{1}$$

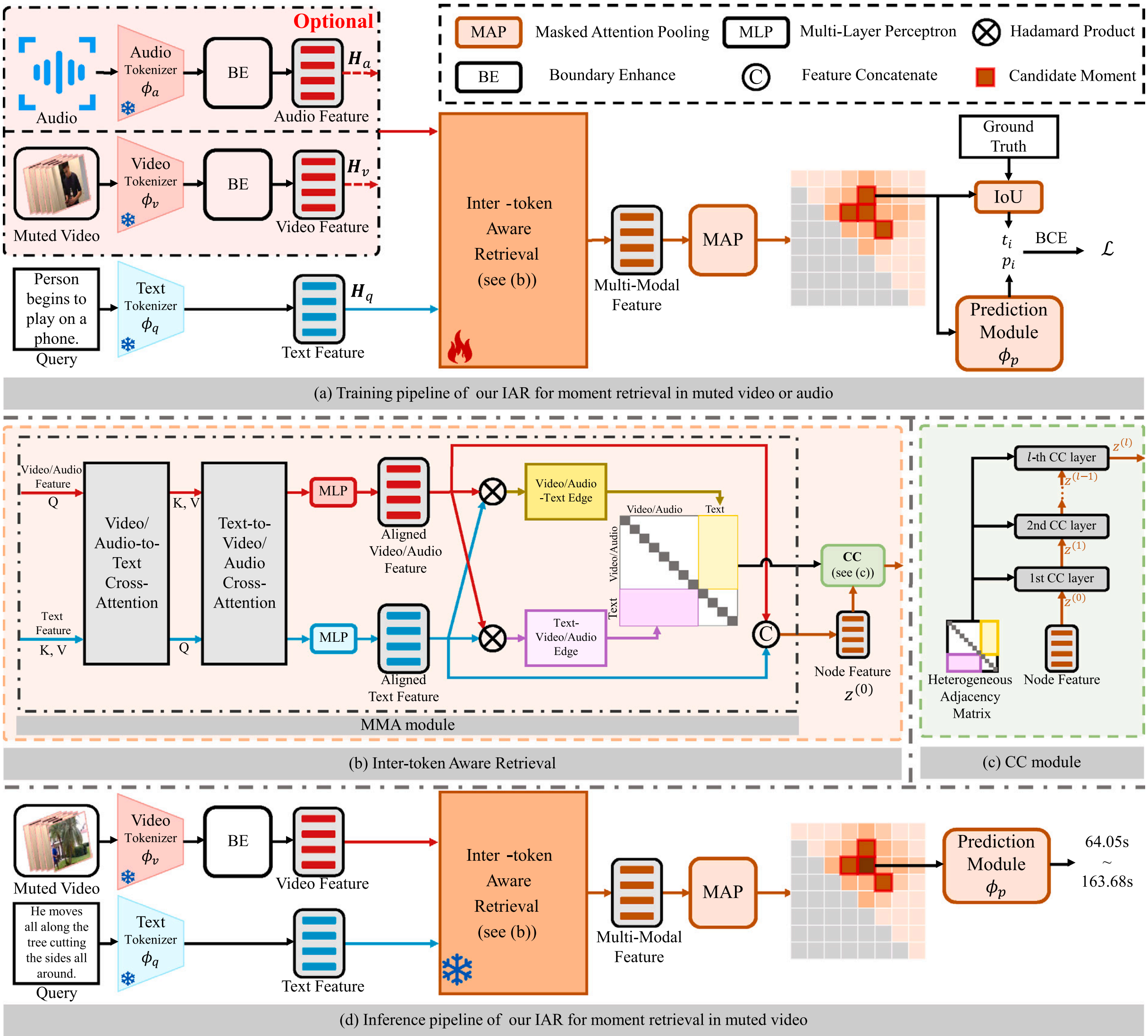


Fig. 2. The overall architecture: (a) The whole training pipeline with feature tokenizers, BE, MMA, and CC. (b) IAR builds a graph and performs multi-modal feature interaction. (c) The CC module. (d) The inference pipeline for moment retrieval in muted video.

where M denotes a muted video or audio, $f(\cdot,\cdot)$ denotes a model consisting of visual/textual/acoustic tokenizers $\phi_v/\phi_a/\phi_q$, cross-modal interaction module ϕ_i, and prediction module ϕ_p.

Unlike multi-modal fusion frameworks focusing on fusion, our IAR is intentionally designed to operate multi-modal fusion under uni-modal constraints of visual or acoustic features, reflecting realistic usage conditions. Its architecture is shown in Fig. 2. Firstly, we input the untrimmed video/audio and query into the feature tokenizer to obtain the initial visual/acoustic features, as well as the textual features. Then, we design ϕ_i, which comprises BE, MMA, and CC, to jointly employ uni-modal and multi-modal interactions in context while preserving the intrinsic temporal boundary characteristics. Finally, the obtained multi-modal features are passed to the prediction module ϕ_p to generate candidate moments with prediction scores. Non-maximum suppression (NMS) is applied to eliminate redundant moments and obtain the ultimate predictions.

3.2. Feature tokenizer

For a sentence query Q, we divide it into L_q tokens $Q = \{q_t\}_{t=1}^{L_q}$ and feed them into a pre-trained text tokenizer to extract text features:

$$\boldsymbol{H}_q = \phi_q(Q) \in \mathbb{R}^{L_q \times d}, \tag{2}$$

where d is the hidden dimension.

For a given untrimmed muted video V, we divide it into L_v tokens $V = \{v_i\}_{i=1}^{L_v}$. Each token v_i contains a fixed number of frames. After that, we use a pre-trained 3D CNN backbone to extract visual features and add 1D positional embeddings for visual tokens as follows:

$$\boldsymbol{X}_v = \phi_v(V) + \gamma(V) \in \mathbb{R}^{L_v \times d}, \tag{3}$$

where $\boldsymbol{X}_v$ represents visual token representations, $\gamma(V)$ denotes positional embeddings of video tokens, $\phi_v(\cdot)$ is the pre-trained 3D CNN backbone, and d is the hidden dimension.

For a given untrimmed audio A, we divide it into L_a tokens $A = \{a_c\}_{c=1}^{L_a}$. After that, we use a Convolutional Recurrent Neural Network (CRNN) [37] to extract audio features and add 1D positional embeddings for audio tokens as follows:

$$\boldsymbol{X}_a = \phi_a(A) + \gamma(A) \in \mathbb{R}^{L_a \times d}, \tag{4}$$

where $\phi_a(\cdot)$ denotes the CRNN backbone, d is the hidden dimension, and $\gamma(A)$ represents positional embeddings for audio tokens.

3.3. Boundary enhance (BE)

With multi-modal interaction, the diversity of visual/acoustic features diminishes, increasing feature similarity between different tokens and blurring their boundaries. To preserve precise boundaries in subsequent computational processes, we propose a BE block to enhance the discernibility of feature boundaries between events and backgrounds. Specifically, inspired by Ref. [38], we increase the feature distance between event and non-event of visual/acoustic tokens by incorporating average features of the whole clip, enabling the model to focus more on capturing boundaries between events and backgrounds while maintaining the rich semantic information of uni-modal features:

$$\begin{aligned}\phi_{BE}(x) = {} & \mathrm{ReLU}\left(\mathrm{FC}\left(\mathrm{AvgPool}(x)\right)\right)\mathrm{FC}(x) \\ & + \mathrm{Conv}_w(x)\left(\mathrm{Conv}_w(x) + \mathrm{Conv}_{kw}(x)\right) + x,\end{aligned} \tag{5}$$

where x denotes input video/audio features $\boldsymbol{X}_v/\boldsymbol{X}_a$, FC and $\mathrm{Conv}_p, p \in \{w, kw\}$ denote the fully-connected layer and 1D depth-wise convolution layer. $\mathrm{AvgPool}(x)$ is the average pooling for all features over the temporal dimension.

We feed $\boldsymbol{X_v}/\boldsymbol{X_a}$ into the BE block and obtain the final visual/audio representations $\boldsymbol{H}_v/\boldsymbol{H}_a$ as follows:

$$\boldsymbol{H}_v = \phi_{BE}(\boldsymbol{X}_v) \in \mathbb{R}^{L_v \times d}, \boldsymbol{H}_a = \phi_{BE}(\boldsymbol{X}_a) \in \mathbb{R}^{L_a \times d}. \tag{6}$$

3.4. Multi-modal alignment (MMA)

In this section, we will introduce MMA. We treat the nodes of visual/acoustic/textual tokens as different types of nodes and construct a graph. Specifically, we employ cross-attention to align different uni-modal features, and then construct edges and obtain a graph adjacency matrix. We build the relationships between different nodes while using their degree of relevance as the weights of the edges. Therefore, we first align visual/acoustic and textual features to strengthen query-relevant information by interacting with different types of node features through the cross-attention block. Then, we construct the corresponding graph where edges capture uni-modal and multi-modal interactions. Lastly, we concatenate these features into an adjacency matrix. Cross-attention is as follows:

$$\boldsymbol{Attn}_{q2m} = \frac{\left(\boldsymbol{H}_q \cdot \boldsymbol{W}^q_{q2m}\right) \cdot \left(\boldsymbol{H}_m \cdot \boldsymbol{W}^m_{q2m}\right)^\top}{\sqrt{d}}, \tag{7}$$

$$\boldsymbol{Attn}_{m2q} = \frac{\left(\boldsymbol{H}_m \cdot \boldsymbol{W}^m_{m2q}\right) \cdot \left(\boldsymbol{H}_q \cdot \boldsymbol{W}^q_{m2q}\right)^\top}{\sqrt{d}}, \tag{8}$$

$$\boldsymbol{F}_q = \mathrm{MLP}_q\left(\mathrm{softmax}(\boldsymbol{Attn}_{q2m}) \cdot \boldsymbol{H}_m\right) \in \mathbb{R}^{L_q \times d}, \tag{9}$$

$$\boldsymbol{F}_m = \mathrm{MLP}_m\left(\mathrm{softmax}(\boldsymbol{Attn}_{m2q}) \cdot \boldsymbol{H}_q\right) \in \mathbb{R}^{L_m \times d}, \tag{10}$$

where the $\boldsymbol{W}^q_{q2\,m}$, $\boldsymbol{W}^m_{q2\,m}$, $\boldsymbol{W}^m_{m2q}$, $\boldsymbol{W}^q_{m2q}$ are trainable weight matrices, MLP_q, MLP_m are the textual query and visual/acoustic Multi-Layer Perceptrons (MLPs). $\boldsymbol{F}_q$, $\boldsymbol{F}_m$ denote the aligned textual query feature and visual/acoustic feature.

Based on the aligned features $\boldsymbol{F}_q$ and $\boldsymbol{F}_m$, we calculate the video/audio-to-text and text-to-video/audio edge weights as shown in Eq. (11) and Eq. (12):

$$\boldsymbol{E}_{m2q} = \sigma\left((\boldsymbol{F}_m)(\boldsymbol{F}_q)^\top\right) \in \mathbb{R}^{L_m \times L_q}, \tag{11}$$

$$\boldsymbol{E}_{q2m} = \sigma\left((\boldsymbol{F}_q)(\boldsymbol{F}_m)^\top\right) \in \mathbb{R}^{L_q \times L_m}, \tag{12}$$

where $\boldsymbol{E}_{m2q}$ and $\boldsymbol{E}_{q2\,m}$ denote video/audio-text edge weights and text-video/audio edge weights, respectively, and σ is a non-linear activation function.

Then we concatenate $E_{q2\ m}$ and E_{m2q} into the graph adjacency matrix $\mathbb{A}$, and add self-connections as follows:

$$\tilde{\mathbb{A}} = \mathbb{A} + \boldsymbol{I} \in \mathbb{R}^{(L_m+L_q)\times(L_m+L_q)}, \tag{13}$$

where $\boldsymbol{I}$ is the node self-connections. We concatenate visual/acoustic features with text features as the initial graph node-level features.

$$\boldsymbol{Z}^{(0)} = \left[\boldsymbol{F}_m; \boldsymbol{F}_q\right] \in \mathbb{R}^{(L_m+L_q)\times d}, \tag{14}$$

where $[;]$ denotes the concatenation operation.

3.5. Context convolution (CC)

After obtaining the initial adjacency matrix $\tilde{\mathbb{A}}$, we design CC to employ uni-modal and multi-modal interactions. CC ascertains the similarity between edges, which updates each visual/acoustic token's relevance to the query dynamically during training:

$$\boldsymbol{Z}^{(l+1)} = f_c\left(\prod_{c=1}^{C} \sigma(\tilde{D}^{-1}\tilde{\mathbb{A}}\boldsymbol{Z}^{(l)}\boldsymbol{W}^{(l)})\right), \tag{15}$$

where C denotes the number of channels, $\tilde{D}$ is the degree matrix of $\tilde{\mathbb{A}}$, $\boldsymbol{W}^{(l)}$ is a trainable weight matrix, $\boldsymbol{Z}^{(l)}$ denotes the node representations from the l-th CC layer, $\boldsymbol{Z}^{(0)} \in \mathbb{R}^{(L_m+L_q)\times d}$ is a feature matrix that transforms $\boldsymbol{H}_v/\boldsymbol{H}_a$ and $\boldsymbol{H}_q$ to the same dimension d and concatenates them. f_c is a channel aggregation function, and σ is a non-linear activation function.

The CC returns an updated graph whose nodes are represented by multi-modal features. The node features are fused with information from the other modality, aggregating similar multi-modal relations while preserving uni-modal relations. Then, we use a masked attention pooling layer to generate candidate moments, following [16].

3.6. Loss function

Following [15], we use t_i as the soft label by scaling the IoU score o_i by two thresholds t_{min} and t_{max}:

$$t_i = \begin{cases} 0 & o_i \le t_{min}, \\ \frac{o_i - t_{min}}{t_{max} - t_{min}} & t_{min} < o_i < t_{max}, \\ 1 & o_i \ge t_{max}, \end{cases} \tag{16}$$

Then, each candidate moment is fed to an MLP to compute the alignment confidence score p_i. Our IAR is trained by a binary cross-entropy loss as:

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n} t_i \log p_i + \left(1 - t_i\right)\log\left(1 - p_i\right), \tag{17}$$

where n denotes the number of candidate moments.

4. Experiment

4.1. Datasets

Charades-STA [7]: It is a large-scale collection of video clips designed to advance research in moment retrieval. These videos capture a wide range of everyday indoor activities such as cooking, eating, and cleaning, typically occurring in home environments. What sets Charades-STA apart is its detailed temporal annotations, which include the start and end times of actions along with their corresponding category labels. Specifically, it comprises 6672 videos and 16,128 moment-query annotation pairs, with the training set utilizing 12,408 pairs and the test set employing 3720 pairs. Each video is annotated with 2.4 moments on average.

ActivityNet-Captions [39]: It encompasses 200 distinct activity categories, ranging from simple individual actions like eating and running to complex multi-person interactions such as team sports and concerts. Each video is annotated with temporal information, including the start and end times of activities, and has an average duration of about 10 min. It is divided into three parts: the training set (train), validation set 1 (val_1), and validation set 2 (val_2), which contain 37,417, 17,505, and 17,031 moment-query pairs, respectively. Following [8], these three parts are used sequentially for model training, validation, and testing. Each video is annotated with more than three moments on average.

AudioGrounding-AMR: The original AudioGrounding database[4] is based on AudioCaps [40], which is established using part of an audio event dataset, AudioSet [41]. AudioSet establishes an expanded ontology system encompassing 632 audio event categories, incorporating 2,084,320 manually annotated 10-second audio clips extracted from YouTube videos. The ontology hierarchically organizes sound events across human/animal sounds, instruments/music, and everyday environments. Consequently, the original AudioGrounding may have multiple ground-truth values for a sentence query. Thus, we reconstruct it by maintaining a single sentence query corresponding to only one ground-truth instance. Specifically, we exclude annotations where one query corresponds to multiple

[4] https://github.com/wsntxxn/TextToAudioGrounding [9]

Table 1
The detail of the experimental environment.

Name	Configuration
Central processing unit	12th Gen Intel(R) Core(TM) i7-12700KF
Graphic processing unit	NVIDIA RTX 3090 × 1
Hard disk	SAMSUNG SSD 980 PRO 2TB
Memory	KINGSTON KF3200C16D4/32GX × 4
Operation system	Ubuntu 22.04.5 LTS
Machine learning framework	PyTorch 1.12.0

ground-truth instances and retain only those annotations where one query corresponds to a single ground-truth instance, constructing the AudioGrounding-AMR. Finally, we have 2933 pairs for training and 540 pairs for testing. Note that we examine the query-length distributions before and after reconstruction and find no significant difference, with only a slight shift in duration. Because the filtering is content-agnostic and relies solely on enforcing a one-to-one query-moment mapping, any selection bias introduced by this step is negligible.

4.2. Evaluation metrics

We adopt two types of metrics: (1) Recall for evaluating the accuracy of retrieval, and (2) Time, FLOPs, and Params for evaluating the efficiency of retrieval.
Recall: We use "R@n,IoU = m" (abbreviated as "Rn@m"), following [7]. It means the percentage of natural language queries with at least one temporal segment in top-n predictions whose IoU with the ground truth is larger than m. For ActivityNet-Captions dataset, Charades-STA dataset, and AudioGrounding-AMR dataset, we set $n = 1$ and $m \in \{0.5, 0.7\}$.
Time, FLOPs, and Params: We adopt "Time", "FLOPs", and "Params" to measure efficiency by combining [37] and [42]. "Time" represents the inference time (milliseconds) from the input of video/audio and query to the prediction output. "FLOPs" represents the floating-point operations, measuring the model's computational costs. "Params" represents the parameter size of the model. Notably, the statistics include word embeddings.

4.3. Implementation details

Feature tokenizer. For video, we use pre-trained C3D [43] for ActivityNet-Captions, and I3D [44] for Charades-STA.

For audio, we adopt CRNN [37]. Note that we use the same settings to re-implement MVMR methods for AMR. For text, we use [45] to extract the token-level features as textual features.
Training and inference settings. We adopt Adam [46] with an initial learning rate of 8e-5, 5e-4, and 4e-5 for ActivityNet-Captions, Charades-STA, and AudioGrounding-AMR, respectively. The batch size is 32, 64, and 32 for three datasets. All the NMS thresholds are 0.5. The IoU thresholds (t_{min}, t_{max}) for the three datasets are: (0.80, 0.81), (0.50, 0.80), (0.50, 0.80). The main experimental environment is shown in Table 1.

4.4. Baseline methods

To compare the efficacy of our IAR model with existing methods, we select more than ten moment retrieval methods as baselines on ActivityNet-Captions and Charades-STA. On our reconstructed AudioGrounding-AMR dataset, we re-implement six methods by replacing the visual encoder with the audio encoder. The six methods include:

- **MS-2D-TAN** [8]: It defines the start and end timestamps of a moment as the horizontal and vertical axes of a two-dimensional space, where each point in this space represents a moment. The distance between points indicates the contextual relationship between moments, while different scales on the axes represent the adjacency of these contextual relationships. This method combines multiple two-dimensional adjacency matrices of varying scales to construct a multi-scale two-dimensional temporal adjacency network.
- **VLG-Net** [16]: It is a graph matching network that independently models short frame sequences of videos and text into nodes. After graph matching, it obtains visual nodes augmented by the text. Finally, potential candidate moments are extracted through a masked attention pooling layer.
- **EMB** [24]: It posits that the start and end timestamps annotated in existing datasets are not deterministic; rather, due to the smooth transition of events within videos, these timestamps carry significant uncertainty. The network proposes a Guided Attention mechanism to model this uncertainty in timestamp information by optimizing the selection of frame-wise endpoints subject to segment-wise content.
- **MomentDiff** [25]: It samples random temporal segments as initial guesses and obtains a mapping from any random position to the true moment through a well-trained diffusion model. It iteratively refines these moments from random locations to generate accurate temporal boundaries.

Table 2
Comparisons of performance on ActivityNet-Captions. The results of comparisons are reported in the original paper. The visual feature encoder is C3D in ActivityNet-Captions, "*" denotes a method that uses another visual feature encoder (e.g., SlowFast [47], Video Swin Transformer [48]). The top-3 results of each dimension (column) are colored red, blue, and green, respectively.

Method	Source	Year	R1@0.5	R1@0.7
SCDM [14]	Neurips	2019	36.75	19.86
2D-TAN [15]	AAAI	2020	44.51	26.54
VLG-Net [16]	ICCV	2021	46.32	29.82
MS-2D-TAN [8]	TPAMI	2021	46.16	29.21
ADPN [11]	ACM MM	2023	39.56	25.20
SGMN [17]	CVIU	2023	47.18	31.77
*ProTéGé [26]	CVPR	2023	45.99	29.02
TCFN [23]	ICME	2023	40.58	24.73
M^{2}DCapsN [18]	TNNLS	2023	47.03	29.99
SEEN [19]	TOIS	2023	45.56	28.10
CCL [20]	WACV	2024	47.73	31.21
IAR(ours)	-	-	**48.46**	**33.12**

- **QD-DETR** [36]: It incorporates cross-attention layers in the initial stages of the Transformer encoder to integrate the context of the text query into the video representation. It modifies the 2D dynamic anchor boxes from object detection methods to represent 1D moments in the video.
- **TR-DETR** [22]: It aligns visual and textual features in a shared latent space using local and global regularization components. Based on the aligned textual features, it filters out query-irrelevant information in visual features.

5. Results

5.1. Performance comparison

We compare the performance of our IAR with some state-of-the-art methods in MVMR/AMR tasks. Note that existing AMR methods are limited. To address this, we re-implement six state-of-the-art MVMR methods based on their open-source implementations using an audio tokenizer to replace the video tokenizer. Regarding structures and hyper-parameters, we refer to their published settings. We evaluate our IAR under identical dataset partitions and standardized feature preprocessing protocols on three benchmark datasets: ActivityNet-Captions, Charades-STA for the MVMR task, and AudioGrounding-AMR for the AMR task. Additionally, our goal is to design a lightweight and efficient architecture to address these two tasks. Hence, we also conduct a comparative analysis of the parameters and computational efficiency of the proposed method against other two-stage approaches.

MVMR performance comparison. For the MVMR task, we demonstrate the performance of our IAR in Table 2 and Table 3, alongside results from multiple baselines in the literature. Experimental results indicate that our IAR exhibits competitive or at least comparable performance relative to current state-of-the-art techniques. Specifically, on ActivityNet-Captions and Charades-STA, IAR outperforms the second-best method by an average of 1.4 % and 5.3 % on R1@0.5 and R1@0.7, respectively.

AMR performance comparison. The experimental results in Table 4 demonstrate that our IAR exhibits strong generalization to the audio domain and outperforms comparisons. Most of the compared methods adopt GloVe as their text tokenizer. To ensure a fair comparison, we implement a variant of our model using GloVe embeddings, denoted as "IAR w/ GloVe." As shown in the results, IAR w/ GloVe significantly outperforms all baseline models, despite using the same textual representation. Interestingly, it even surpasses the original IAR model equipped with BERT. A plausible explanation is that queries in the AudioGrounding-AMR dataset are relatively simple, so lightweight embeddings, such as GloVe, exhibit a lower effective rank than representations from more complex pre-trained models like BERT [49]. We compute the effective rank r_{eff} and the token-independence index (IND) for text features extracted by BERT and GloVe. r_{eff} is computed from the feature matrix $\boldsymbol{H}_q$ by mean-centering along tokens, and taking singular values. IND is computed by constructing a cosine similarity matrix of token embeddings. We define IND as the mean absolute off-diagonal entry divided by the mean diagonal entry. Smaller values indicate greater independence. GloVe yields significantly lower effective rank (2.63 vs. 4.89; non-overlapping 95 % CIs) and lower token-independence index (0.215 vs. 0.525; non-overlapping 95 % CIs), indicating sharper, lower-rank and more independent token features. Although variants that employ different text tokenizers exhibit performance differences, these differences remain modest, which further demonstrates the robustness and effectiveness of IAR.

Table 3
Comparisons of performance on Charades-STA. The results of comparisons are reported in the original paper. The visual feature encoder is I3D in Charades-STA, "*" denotes a method that uses another visual feature encoder (e.g., SlowFast [47], Video Swin Transformer [48]). The top-3 results of each dimension (column) are colored red, blue, and green, respectively.

Method	Source	Year	R1@0.5	R1@0.7
SCDM [14]	NeurIPS	2019	54.44	33.43
VSLNet [35]	ACL	2020	47.31	30.19
MS-2D-TAN [8]	TPAMI	2021	56.64	36.21
ADPN [11]	ACM MM	2023	55.32	37.47
QD-DETR [36]	CVPR	2023	50.67	31.02
*ProTéGé [26]	CVPR	2023	53.26	30.38
*LLaViLo [27]	ICCV	2023	55.72	33.43
TCFN [23]	ICME	2023	51.77	29.95
MomentDiff [25]	NIPS	2023	55.57	32.42
M^2DCapsN [18]	TNNLS	2023	55.03	31.61
SEEN [19]	TOIS	2023	54.11	33.44
TR-DETR [22]	AAAI	2024	55.51	33.66
IAR(ours)	-	-	**57.37**	**39.78**

Table 4
Comparisons of performance on AudioGrounding-AMR dataset. The top-2 results of each dimension (column) are colored red and blue.

Dataset	AudioGrounding-AMR			
Method	**Source**	**Year**	**R1@0.5**	**R1@0.7**
MS-2D-TAN* [8]	TPAMI	2021	50.74	40.19
VLG-Net* [16]	ICCV	2021	52.04	42.96
EMB* [24]	ECCV	2022	50.56	43.89
MomentDiff* [25]	NeurIPS	2023	24.34	16.23
QD-DETR* [36]	CVPR	2023	33.67	23.73
TR-DETR* [22]	AAAI	2024	32.05	18.86
IAR w/ Glove	-	-	**55.19**	**46.30**
IAR(ours)	-	-	**54.07**	**45.19**

* indicates that these methods are designed for the MVMR task; we replaced the video tokenizer with an audio tokenizer and then re-implemented it for the AMR task.

Parameters and efficiency comparison. We compare our IAR with classic two-stage methods in efficiency on Charades-STA and ActivityNet-Captions in terms of Time, FLOPs, and Params (Table 5). We summarize the statistics for each two-stage model based on its official open-source implementation code. Besides, we evaluate the performance of every model in R1@0.7 (termed as "ACC@1"). Our IAR achieves the best performance on ActivityNet-Captions, and even surpasses CCA, an acceleration technique specifically optimized for the VMR task. We also report the results of two variants, IAR-128 and IAR-256, where d is changed to 128 and 256, respectively. The results indicate that we can further improve speed while maintaining performance by reducing feature dimension.

5.2. Ablation study

In this section, we perform ablation studies on these aspects: (1) effects of different components in our IAR, (2) mechanism of BE, (3) effects of various weight calculation methods when constructing graphs in MMA, (4) effects of the edge aggregation in CC, and (5) effects of IoU thresholds t_{min} and t_{max} in Eq. (16).

Table 5
Overall comparisons on Time (ms), FLOPs ($\times 10^8$), Parameters ($\times 10^6$), and Accuracy (R1@0.7). "IAR-128" and "IAR-256" are variants of IAR with different d. All the results are estimated in the same test case. The best performance of two-stage methods is in **bold**.

Method	Charades-STA				ActivityNet-Captions			
	Time ↓	FLOPs ↓	Params ↓	Acc@1 ↑	Time ↓	FLOPs ↓	Params ↓	Acc@1 ↑
Two-stage methods								
2D-TAN [15]	6.41[1]	521.95[1]	60.93[1]	23.31[1]	13.88	4985.21	91.59	26.54
MS-2D-TAN [8]	9.66	4780.99	477.89	36.21	3.29	2732.30	479.46	29.21
CCA [50]	–[2]	–[2]	–[2]	–[2]	8.81	1371.61	79.50	29.37
IAR(ours)	**2.59**	**259.40**	**26.79**	**39.78**	**1.73**	**39.81**	**15.46**	**33.12**
Variants								
IAR-128	1.31	16.32	1.94	34.52	1.22	3.12	1.79	28.41
IAR-256	1.58	64.96	6.97	36.80	1.36	10.50	4.50	29.51

[1] This means the results are reported using VGG visual features, which differ from our I3D features.
[2] This means the original paper does not include this metric's performance or a pre-trained model.

Table 6
Ablation study on the components of IAR.

	BE	MMA	CC	Charades-STA		ActivityNet-Captions		AudioGrounding-AMR	
				R1@0.5 ↑	R1@0.7 ↑	R1@0.5 ↑	R1@0.7 ↑	R1@0.5 ↑	R1@0.7 ↑
(a)				30.86	16.72	24.10	13.11	47.96	41.11
(b)	✓			34.25	20.56	28.29	14.55	52.04	43.89
(c)		✓		53.41	34.11	46.10	29.56	52.41	43.89
(d)			✓	42.39	25.91	40.28	25.40	53.15	44.26
(e)	✓	✓		51.42	32.42	46.05	28.62	52.22	43.70
(f)	✓		✓	42.26	26.24	46.09	30.00	51.30	42.04
(g)		✓	✓	56.53	36.72	46.44	29.45	53.70	44.26
(h)	✓	✓	✓	**57.37**	**39.78**	**48.46**	**33.12**	**54.07**	**45.19**

Table 7
Average rank of visual features from each module on Charades-STA and AudioGrounding-AMR.

Dataset	Method	Input	BE	MMA	CC
Charades-STA	IAR w/o BE	32.0	–	14.7	32.0
	IAR	32.0	32.0	11.8	32.0
AudioGrounding-AMR	IAR w/o BE	63.9	–	25.1	64.0
	IAR	63.9	64	17.9	64.0

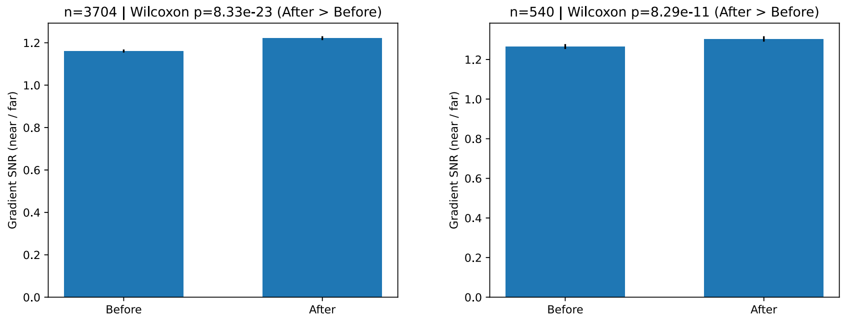
5.2.1. Effects of the components

The contributions of the different components are investigated on Activity-Captions, Charades-STA, and AudioGrounding-AMR datasets. As shown in Table 6, the effects of BE, MMA, and CC are evaluated in terms of "R1@0.5" and "R1@0.7". Compared to the baseline (a), rows (b) to (d) show the efficacy of each component and verify that each module contributes to the performance improvement of our IAR. BE can efficiently enhance the boundaries of video/audio and enable models to be more accurately grounded (e.g., (a,b) and (g,h)). BE boosts the performance when combined with both MMA and CC, although it declines when only combining BE and MMA. Notably, (h) shows improvements over (g) in R1@0.7 by 8.3 %, 12.5 %, and 2.1 % across three datasets, significantly outperforming the R1@0.5 improvements. The stricter boundary criteria of R1@0.7 likely better highlight the effectiveness of BE. It also demonstrates the importance of preserving the uni-modal temporal context during multi-modal fusion. The efficacy of MMA demonstrates the significance of computing video/audio tokens' relevance to the query in context (compare between (f) and (h)). The efficacy of CC shows that employing uni-modal and multi-modal interactions is significant (compare between (e) and (h)). In sum, our components complement one another, and our motivation is verified.

5.2.2. Mechanism of BE

Our BE module can distinguish the boundary information of videos and audio while maintaining their feature diversity as much as possible. We confirm this by computing the rank of visual/acoustic features in our IAR and its variant, IAR w/o BE, on Charades-STA and AudioGrounding-AMR. The results are shown in Table 7. The model employing BE exhibits a decline in feature rank following MMA, indicating that BE tends to classify features into foreground and background during alignment. The rank is restored after applying CC, suggesting that the diversity of features is preserved.

Complementing the rank-based evidence, we further quantify boundary sensitivity by calculating gradient signal-to-noise ratio (SNR), as shown in Fig. 3. On Charades-STA, BE significantly increases the boundary gradient SNR (Wilcoxon one-tailed $p < 0.05$). On AudioGrounding-AMR, BE also yields a significant improvement ($p < 0.05$). In both datasets, the mean SNR remains above one



(a) Charades (b) AudioGrounding-AMR

Fig. 3. Boundary gradient signal-to-noise ratio (SNR, near / far) before and after the BE module on (a) Charades-STA and (b) AudioGrounding-AMR. Bars show the mean across samples. BE increases SNR at annotated boundaries on both datasets.

Table 8
Ablation study on the construction of edges. “SMM” denotes replacing multi-head attention with matrix multiplication. “COS” denotes replacing multi-head attention with cosine similarity. The results are reported with R1@0.5 and R1@0.7.

Method	ActivityNet-Captions		AudioGrounding-AMR	
	R1@0.5 ↑	R1@0.7 ↑	R1@0.5 ↑	R1@0.7 ↑
SMM	46.36	30.78	52.41	43.89
COS	45.66	30.25	52.59	43.89
IAR (w/ MMA)	**48.46**	**33.12**	**54.07**	**45.19**

and increases after BE, indicating that BE consistently sharpens temporal changes at boundaries while suppressing off-boundary noise across video and audio modalities.

5.2.3. Effects of MMA

We validate the effects of the cross-attention in MMA on the edge construction in the following ways: 1) we compare IAR with variants whose cross-attention is replaced by simple matrix multiplication (denoted as “SMM”) and cosine similarity (denoted as “COS”). As shown in Table 8, MMA utilizes the cross-attention mechanism to compute the attention scores between the video/audio tokens and query tokens, thereby performing multi-modal feature alignment before interaction. The attention scores serve as edge weights, while MMA enables learning for edges, facilitating the creation of a large multi-modal representation space. In contrast, COS and SMM compute similarity only for unaligned features.

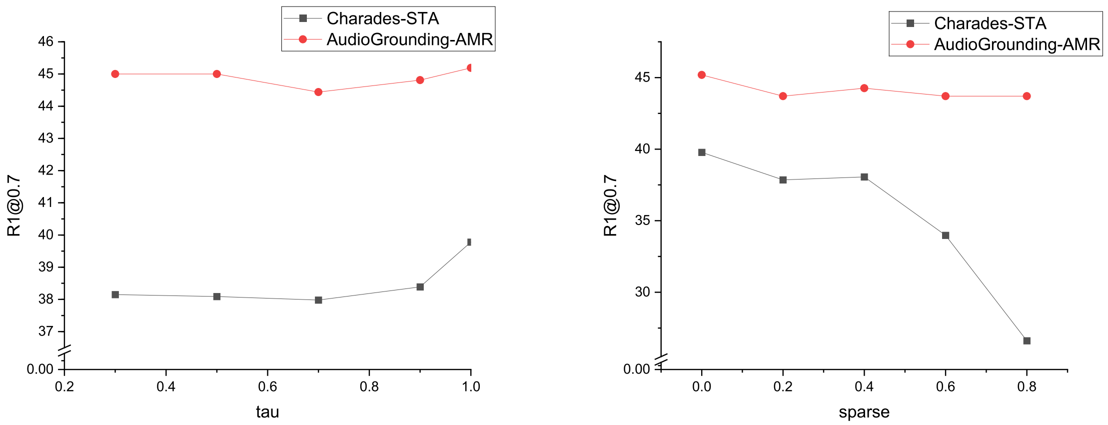
We next examine the sensitivity of this mechanism to the temperature τ and the sparsification threshold t_{sparse}. We evaluate $\tau \in \{1.0, 0.9, 0.7, 0.5, 0.3\}$ and $t_{sparse} \in \{0, 0.2, 0.4, 0.6, 0.8\}$ on Charades-STA for MVMR and AudioGrounding-AMR for AMR, respectively. The results in Fig. 4(a) show that both datasets reach their highest R@1@0.7 when $\tau = 1.0$. Lower temperatures yield flatter or worse curves, suggesting that over-sharpened attention under-utilizes contextual evidence, whereas $\tau = 1.0$ maintains an effective balance between focus and coverage. Fig. 4(b) illustrates performance degradation as sparsification increases: on Charades-STA, the drop is pronounced, and on AudioGrounding-AMR, the decline is mild but consistent. The best results occur at $t_{sparse} = 0$, indicating that pruning low-weight edges removes useful contextual links and harms retrieval. We therefore adopt $\tau = 1.0$ and $t_{sparse} = 0$ as the default configuration.

5.2.4. Effects of the edge aggregation

In our IAR, we aggregate only the multi-modal edges. We compare our IAR with variants that aggregate uni-modal edges only or merge uni- and multi-modal edges. Results in Table 9 show that our IAR performs better than other variants. When we use only uni-modal edges, the performance on the Charades-STA, ActivityNet-Captions, and AudioGrounding-AMR datasets drops by at least 5.12 %. The decline is attributed to insufficient multi-modal interaction. Incorporating uni-modal edges with multi-modal edges is less effective than multi-modal edges alone, as updating uni-modal edges partially disrupts the temporal uni-modal context. Additionally, on the AudioGrounding-AMR dataset, the three variants show similar performance, possibly because audio features are less semantically rich, resulting in less boundary blurring with uni-modal edges.

5.2.5. Effects of t_{min} and t_{max}

We analyze the effects of t_{min} and t_{max} on AudioGrounding-AMR, as it is newly created and has no prior work to reference or benchmark against. We also analyze them on Charades-STA for comparison. We evaluate $t_{min} \in \{0.5, 0.55, 0.6\}$ and $t_{max} \in \{0.7, 0.75, 0.8\}$,

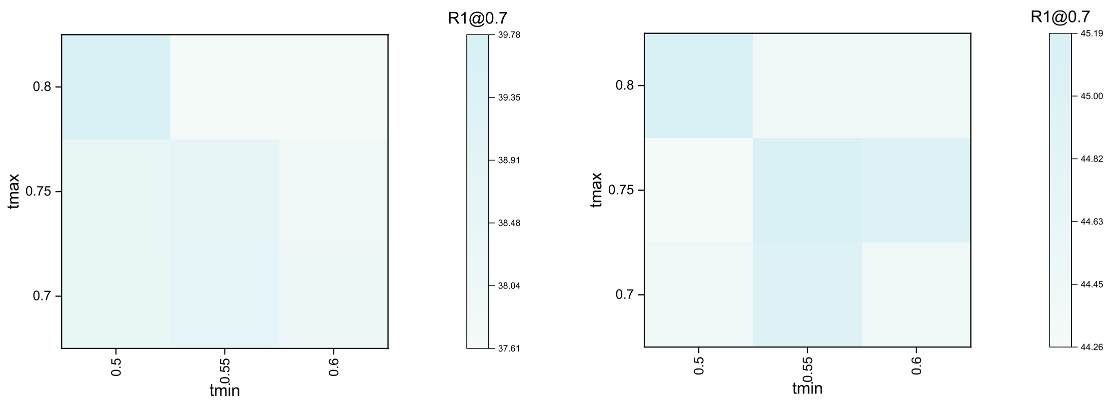


(a) Temperature parameter   (b) Sparsification thresholding

Fig. 4. Simple effects of (a) attention temperature and (b) sparsification R1@0.7. Curves are shown for Charades-STA (gray) and AudioGrounding-AMR (red).

Table 9
Ablation study on edge aggregation. The results are reported with R1@0.5 and R1@0.7. "MM" denotes using multi-modal edges. while "UM" denotes using uni-modal edges.

MM	UM	Charades-STA		ActivityNet-Captions		AudioGrounding-AMR	
		R1@0.5 ↑	R1@0.7 ↑	R1@0.5 ↑	R1@0.7 ↑	R1@0.5 ↑	R1@0.7 ↑
	✓	36.34	22.47	28.29	14.55	51.30	42.59
✓		**57.37**	**39.78**	**48.46**	**33.12**	**54.07**	**45.19**
✓	✓	56.42	36.91	46.09	29.49	52.78	43.15



(a) Charades   (b) AudioGrounding-AMR

Fig. 5. The heatmap of R1@0.7 over t_{min} and t_{max} on Charades and AudioGrounding-AMR.

and report R1@0.7 on these two datasets, as shown in Fig. 5. Within $t_{min} \in [0.5, 0.6]$ and $t_{max} \in [0.7, 0.8]$, the variation of R1@0.7 is small (coefficient of variation 1.76 %/0.84 %, range-to-mean 5.65 %/2.08 %).

We also perform linear regression and Spearman's rank tests. For Charades-STA, the tests show a mild, non-significant decreasing trend with t_{min} ($\rho \approx -0.58, p = 0.12$) and a negligible effect of t_{max} ($\rho \approx -0.29, p = 0.45$). For AudioGrounding-AMR, the tests indicate no sensitivity to t_{min} ($\rho \approx 0.03, p = 0.96$) and only a weak, non-significant monotonic increase with t_{max} ($\rho \approx 0.56, p = 0.15$). Therefore, within this hyperparameter grid, R1@0.7 remains insensitive and stable with respect to t_{max} and t_{min}.

5.3. Robustness test

To further demonstrate the robustness of our method, we conduct multiple runs on the AudioGrounding-AMR dataset, calculating the average and standard deviation of the results, as shown in Table 10. By comparing our IAR with the TR-DETR and QD-DETR

Table 10
Comparisons of robust test results on AudioGrounding-AMR dataset. The best result of each dimension (column) is colored red. The values within parentheses represent the standard deviation over 10 runs.

Dataset	AudioGrounding-AMR			
Method	**Source**	**Year**	**R1@0.5**	**R1@0.7**
QD-DETR* [36]	CVPR	2023	31.74 (1.71)	22.74 (1.65)
TR-DETR* [22]	AAAI	2024	24.71 (10.29)	11.42 (7.91)
IAR(ours)	-	-	**53.54 (0.33)**	**44.98 (0.24)**

* indicates that these methods are designed for the MVMR task; we replaced the video tokenizer with an audio tokenizer and then re-implemented it for the AMR task.

Table 11
Analysis of short-/long-query sets of ActivityNet-Captions. The results are reported with R1@0.5 and R1@0.7.

Method	Year	short		long	
		R1@0.5 ↑	R1@0.7 ↑	R1@0.5 ↑	R1@0.7 ↑
VLG-Net	2021	44.63	27.80	47.37	31.05
CCA	2024	45.14	28.17	47.80	29.46
IAR	–	**45.91**	**30.54**	**49.76**	**34.25**

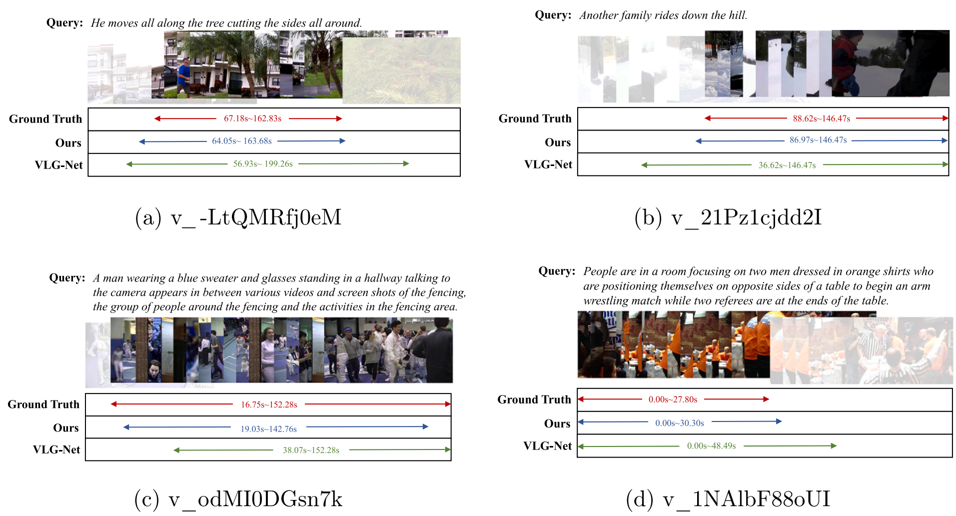


Fig. 6. Examples of retrieval results. We compare our IAR with VLG-Net in ActivityNet-Captions dataset. (a) and (b) serve as an exemplar for short queries, while (c) and (d) serve for long queries.

methods, we have evidenced the effectiveness and robustness of our approach. ANOVA analysis indicates that the leading effect of our IAR is significant.

5.4. Qualitative analysis

We compare IAR with a classical graph-based method, VLG-Net, and a recent method, CCA, on the ActivityNet-Captions test set, which we partition into short-query (6500 samples) and long-query (10,531 samples) subsets using the average query length of 12.74 words as the threshold. Table 11 shows that our IAR outperforms competitors in different query cases. We show four qualitative retrieval results from ActivityNet-Captions in Fig. 6. The top two are from the short-query set, and the bottom two are from the long-query set with 40 words. Short queries may bring unclear descriptions and fuzzy boundaries, while long queries provide rich information but also introduce additional noise. In both cases, our IAR's predictions closely align with the ground truth, surpassing VLG-Net. Especially in Fig. 6b, our IAR performs better on a query with a keyword "another" indicating that our model maintains the temporal uni-modal context while modeling multi-modal interactions.

6. Conclusion and future work

In this paper, we unify the experimental settings for MVMR and AMR and propose a single framework, IAR, that effectively handles both tasks under uni-modal constraints. IAR explicitly models token-level interactions across modalities while preserving the temporal continuity of uni-modal inputs. This token-aware design enables seamless switching between video and audio modalities without requiring separate models for different user groups. Extensive experiments on three datasets and two tasks show consistent gains over existing methods. Notably, IAR maintains high computational efficiency with fewer parameters and faster inference compared to conventional two-stage approaches, making it well-suited for practical deployment. Furthermore, IAR exhibits strong cross-task generalization, achieving stable performance when transferred between MVMR and AMR retrieval tasks. Moreover, the advances presented in this paper now enable us to enlarge the AudioGrounding-AMR dataset to further aid the community.

CRediT authorship contribution statement

Guolong Wang: Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Xun Tu:** Writing – original draft, Software, Resources, Project administration, Data curation. **Sutian Hou:** Writing – review & editing, Methodology. **Yifei Cao:** Writing – review & editing. **Ying Lin:** Writing – review & editing. **Yu Liu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available upon request.

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